

Distributed Classification of SO₂ Emissions: Volcanic vs. Industrial Sources

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Abstract

Accurate classification of SO₂ emissions as volcanic or industrial in origin is critical for environmental monitoring, health policy, and regulatory compliance. This paper presents a distributed PySpark pipeline deployed on Google Cloud Platform Dataproc for pixel-level SO₂ source attribution at scale, integrating TROPOMI satellite observations, ERA5 850 hPa wind back-trajectory correction, and the Fioletov et al. (2023) global source catalogue as ground truth. Applied to a five-day pilot dataset of 13.6 million globally distributed pixels from June 2023, Gradient Boosted Trees and Random Forest both achieve accuracy of 0.91 and F1 of 0.93 under source-level holdout, substantially outperforming a Logistic Regression baseline. ERA5 wind correction improves precision by 0.026 to 0.028 across both models, providing the first quantitative measurement of this effect in the literature. Strong scaling experiments across 2 to 16 workers achieve a 13.4x speedup, reducing a near 7-hour workload to just over 30 minutes, while weak scaling shows only a 29% rise in normalised runtime for an 8-fold increase in data size. Per-stage analysis identifies GBT training as the primary bottleneck under Amdahl's Law, while spatial join stages scale close to linearly.

Keywords

distributed computing, SO₂ source attribution, TROPOMI, PySpark, Gradient Boosted Trees, satellite remote sensing, strong scaling, weak scaling, ERA5 wind correction

1 Introduction

In a field previously dominated by ground-based monitoring instruments, recent advances in satellite remote sensing have enabled high spatial resolution detection of Sulphur Dioxide (SO₂) emissions at a planetary scale [6]. This new capability frontier generates increasingly high volumes and velocity of data, significantly enhancing our cognisance of SO₂ emissions in near real time. The Sentinel-5P TROPOMI instrument generates over 23GB of raw SO₂ and NO₂ observations daily; when combined with hourly ERA5 reanalysis wind fields across a monthly study period, the resulting dataset exceeds 700GB of raw input — a scale which motivates the use of distributed processing frameworks.

Accurately monitoring and reporting on our patterns of emissions is critical to development; as an atmospheric pollutant, SO₂ is hazardous both to human health and to our environment, causing respiratory issues and acid rain [12, 15]. SO₂ emissions can be categorised into anthropogenic and volcanic sources, thus, accurate classification of SO₂ plumes may help us to decipher our progress in reducing emissions and mitigating risks in line with health policy and environmental regulatory compliance.

This report aims to address a gap in the literature: binary classification of SO₂ plumes of anthropogenic or volcanic origin, using Fioletov et al. [7]'s definitive global catalogue of large anthropogenic and volcanic SO₂ sources as ground truth for global implementation. By presenting a distributed PySpark pipeline deployed on Google Cloud Platform (GCP) Dataproc, this report demonstrates that SO₂ source attribution methodology can be scaled to cross-dataset (TROPOMI, ERA5) and cross-cloud (AWS S3 bucket, Google Cloud Storage) integration, enabling pixel-level classification across 13.6 million globally distributed observations.

We conclude that (i) our source-type classification achieves AUC-ROC 0.94–0.96 under source-level holdout using satellite-derived features alone, (ii) ERA5 850 hPa wind back-trajectory correction, applied following Fioletov et al. [7]'s methodology, degrades volcanic source recall by up to 10 percentage points compared to raw-position labelling — the first quantitative measurement of this effect in the literature, (iii) the pipeline exhibits a 13.4x speedup scaling from 2 to 16 workers across distributed scaling experiments, with the SO₂/NO₂ co-location join identified as the primary computational bottleneck.

2 Related Work

Machine learning algorithms have become a widely adopted approach for automatically processing satellite data across environmental and geophysical applications, enabling faster and more scalable analysis. For example, Abdi [1] demonstrated the effectiveness of multiple classifiers on multispectral satellite imagery for land cover mapping in complex boreal landscapes. While Corradino et al. [5] applied supervised machine learning to detect volcanic thermal features in high spatial resolution satellite sensors, including hot fumaroles, degassing and effusive activity, establishing the viability of ML pipelines for TROPOMI-adjacent remote sensing tasks.

Among ensemble methods, Random Forest (RF; Breiman 3) has emerged as a robust baseline for satellite classification due to its resistance to overfitting and natural handling of feature interactions without scaling. Corradino et al. [6] applied an RF model to TROPOMI data for near real-time volcanic SO₂ plume identification across 2019–2023, subsequently integrating Cloud Top Height retrievals for SO₂ mass quantification.

Gradient Boosted Trees (GBT) another popular machine learning algorithm in this area, introduced by Friedman [9] as stage-wise additive expansion in function space. Lawrence et al. [14] demonstrated that stochastic gradient boosting reduces over-fitting on satellite image classification. Although the Extreme Gradient Boosting algorithm (XGBoost) introduced by Chen and Guestrin [4] is used a lot in the experiments of air quality forecasting [17], it is not natively supported in the Spark MLlib distributed framework

used in this study, making the standard GBT Classifier the appropriate choice for our GCP Dataproc pipeline. We additionally include Logistic Regression as a linear baseline to quantify the gain from non-linear ensemble methods and to assess whether the classification boundary is approximately linear in the engineered feature space.

Notably, Corradino et al. [6] also evaluate the Radius Classifier as a comparison, which is a distance-based method assigning labels based on neighbourhood density. It suits well in spatially compact plume signatures, but is less interpretable and harder to scale to the million-level pixel dataset used in this research. Our work extends this line of research by incorporating ERA5 wind back-trajectory correction into the labelling and feature engineering stages, providing a richer atmospheric context for the classifier and enabling a controlled ablation of the wind correction’s contribution to classification performance.

Distributed scaling behaviour is commonly analysed through two frameworks. Amdahl’s Law [2] predicts that serial portions of a program limit the maximum speedup achievable through parallelism, while Gustafson’s Law [10] shows that scaling problem size alongside worker count can keep runtime roughly constant. Végh [23] further refines these models for modern distributed systems, noting that real-world results often deviate due to communication overhead. For shuffle-heavy stages, partition management is a key performance factor: Lu et al. [16] show that partition optimisation in Spark clusters can significantly reduce scheduling overhead, directly relevant to the partitioning strategy discussed later in the report.

3 Data Selection

3.1 Multi-Satellite Air Quality Sulfur Dioxide (SO₂) Database Long-Term L4 Global V2

Fioletov et al. [7]’s definitive version 2 global catalogue of large anthropogenic and volcanic SO₂ sources calculated that degassing volcanoes contribute between one-quarter and one-third of total SO₂ emission, and further quantified historic emission profiles categorised by both anthropogenic emission source type and geographic region. This paper’s methodology utilised manual expert classification of 759 significant emission sources, critically used as the ground truth in this report for classification.

3.2 TROPOMI

Sentinel-5P Level 2 SO₂ and NO₂ products captured by TROPOMI — Tropospheric Monitoring Instrument — are the frontier of SO₂ satellite detection, offering significant reductions in the signal to noise ratio when compared with other instruments such as the Ozone Monitoring Instrument (OMI) and Ozone Mapping Profiling Suite (OMPS) [8].

There is further merit in selecting TROPOMI products; they are stored as netCDF4 granules on a public AWS S3 bucket in eu-central-1, thus in a location which is convenient for distributed computing. The characteristics of the data are as follows: each orbit produces one granule of 100–200 MB containing a 3D array (1 × scanlines × ground pixels) with ~1.9 million pixels per granule — there are 14 orbits per day, motivating the use of distributed computing for processing.

3.3 ERA5

The primary method for handling wind driven plume displacement in pollutant monitoring is back-trajectory analysis (BTA). BTA specifically accounts for a range of meteorological dynamics that impact particulate plume displacement such as advection, horizontal and vertical wind mixing and dry and wet deposition [21]. Its inclusion has become especially common in pollutant classification with the release of the HYSPLIT model by Stein et al. [21], which established the newest version of the methodology and provided globally gridded weather forecast data to pair with observational metrics [19]. The report incorporated BTA into the findings, but in line with the methods used in Fioletov et al. [7] relies instead on the turbulent velocity data provided by ERA5. ERA5 is an effective dataset as it not only provides the u and v wind components needed for BTA; it also specifically reports these metrics at the 850 hPa pressure level where SO₂ and NO₂ reside in the atmosphere [24]. Furthermore, ERA5 also reports its data at an hourly temporal resolution, allowing us to match the wind conditions to the exact satellite overpass timing in comparison to daily average metrics like the TM5 wind-model included in TROPOMI. Lastly, ERA5 BTA was also used by Fioletov et al. [7] in their source emission catalogue, thus incorporating the same techniques into the MSAQSO₂L4 dataset ensures the quality of the data fed into the pipeline matches the existing ground-truth analysis. A limitation of this method is that ERA5’s 28 km resolution mismatches TROPOMI’s 3.5 × 7 km and can only be fixed by merging the datasets to the nearest ERA5 grid via 0.25° rounding. This introduces a ~14 km maximum spatial attribution error, however, in comparison to the 1-day residence time assumption to enable BTA is largely insignificant.

4 Data Ingestion and Preprocessing Pipeline

The pipeline ingests data from three sources spanning two cloud providers. TROPOMI Level 2 SO₂ and NO₂ products are stored as netCDF4 granules on a public AWS S3 bucket (s3://meeo-s5p) in eu-central-1. Across 30 days of June 2023, 833 granules totalling 692 GB were processed, with quality filtering (qa ≥ 0.5 for SO₂, qa ≥ 0.75 for NO₂, cloud fraction < 0.3, each following Fioletov et al. [7]’s preprocessing methodology) retaining 532 million pixel-level observations. Schema inspection of the raw granules during pipeline development revealed multi-height SO₂ vertical column density retrievals at assumed plume altitudes of 1 km and 7 km, which were retained alongside aerosol index, surface altitude, and viewing geometry — the ratio of the 7 km to 1 km retrieval encodes emission altitude and serves as a physical discriminator between volcanic and industrial sources. ERA5 reanalysis wind fields (u and v components at 850 hPa) were extracted from the ARCO ERA5 Zarr store on Google Cloud Storage (gs://gcp-public-data-arco-era5), yielding 747 million hourly grid points at 0.25° resolution across the study period. The 850 hPa pressure level (~1.5 km altitude) was selected to represent boundary-layer transport of industrial SO₂, consistent with the methodology of Fioletov et al. [7]; selecting a single pressure level reduces the ERA5 extraction volume by a factor of 37.

Spark cannot read netCDF4 natively; the format requires HDF5 C libraries on every worker node and Python UDFs that bypass Spark’s Catalyst query optimiser. We therefore adopted a two-stage

architecture that decouples I/O-bound format conversion from compute-bound distributed analytics. Stage 1 runs on a single GCP VM (e2-standard-8 in europe-west1) using 16 parallel workers via Python’s ProcessPoolExecutor, each downloading one granule from S3 with unsigned access, parsing it with netCDF4-python, applying quality filters, and writing Snappy-compressed Parquet to GCS. The script is idempotent — existing output files are detected and skipped on rerun, enabling fault-tolerant incremental processing. This preprocessing reduced 692 GB of raw netCDF4 to 36 GB of Parquet (19× compression), driven by selective column retention (15 of ~60 variables per granule) and columnar compression exploiting spatial correlation between adjacent pixel values. The one-time preprocessing cost of approximately 80 minutes is amortised across all subsequent Spark jobs, which read native Parquet with full predicate pushdown and Catalyst optimisation. Cross-cloud data transfer from AWS Frankfurt to GCS Belgium (sub-50 ms latency, ~300 km) is deliberately concentrated in this preprocessing stage, where it is parallelisable and executed once; ERA5 data resides natively on GCS, requiring no cross-cloud transfer.

An alternative approach — reading netCDF4 directly within Spark executors — was rejected due to native library dependency fragility across heterogeneous cluster nodes, doubled memory usage from in-memory parsing, and the inability to leverage columnar query optimisation. Google Earth Engine, which hosts pre-ingested TROPOMI data, was also rejected as it abstracts distributed computing internals, preventing measurement of scaling behaviour and join strategies that are core deliverables of this study.

5 Feature Engineering

Feature engineering was performed on Spark’s distributed computing API on GCP across a 5-day sample of the raw dataset from June 1st to 5th, 2023. In total, the final dataset held 13.6 million rows of volcanic and industrial emission pixels, each with 14 input features and 1 nearest source name meta-label (see Table 1).

Two key steps were taken to produce this dataset and are as follows:

5.1 SO₂ and NO₂ Co-location

The first key step taken was attributing each SO₂ and NO₂ observation pixel to those sharing the same date, latitude, and longitude to later produce the SO₂ to NO₂ ratio feature. This was achieved by simplifying each pixel’s coordinate to a 0.1° grid (~11 km) by rounding their coordinate to the nearest tenths of a degree — labelled lat_key and lon_key. This creates a standardised grid where each SO₂ and NO₂ pixel can be paired to those sharing the same grid cell and date via a Spark sort-merge join. This introduces an ~15 km misattribution error due to the rounding creating edge cases for pixels on opposite sides of a grid boundary — i.e. SO₂ at 0.149° and 0.151° or ~0.22 km apart will not match but SO₂ at 0.51° and 0.149° will or ~11 km apart. Ultimately, the TROPOMI satellite has a pixel footprint of 3.5 by 7 km, meaning the misattribution range already sits close to its pixels’ spatial aggregation (~2 pixel widths). Furthermore, the probability of a pixel straddling a grid boundary is extremely rare, only 25% will fall within 0.05°, and then its paired pixel would have to share all the same location data and be on the opposite side of said boundary. As a result, this method not only

reduces peak YARN memory from ~600 GB with a nearest pixel approach to 35 GB, but most also finds the same co-location result for most pixels.

5.2 Join to Fioletov et al. [7] Catalog

The combined ERA5 and TROPOMI pixels were attributed within a 150 km distance to one of the 759 point sources from Fioletov et al. [7] catalog using a broadcast join and user-defined vectorised haversine calculation. The broadcast join was chosen as, without a natural join key between the datasets, a standard pythonic search and calculate loop would involve shuffling 160 million candidate pixel rows across the network — approximately 14 GB of payload. The broadcast join reduces the network transfer ~18,000 folds by reading the 86 KB catalog into memory and distributing it to every executor simultaneously (774 KB total payload) — removing the need for inter-executor communication. Once each executor receives a partition, it creates column vectorised latitude and longitude pixel source differences, which are converted into haversine distance vectors ($N_{\text{pixels}}, 759$) where $\text{Dist}_{i,j}$ is the longest km circle difference between TROPOMI pixel i and j . For N pixels per batch this generates four $N \times 759$ single distance and coordinate matrices, totaling at ~2.43 GB per executor, and negates Python overhead with basic linear algebra subprograms saving ~8 minutes per executor.

6 Machine Learning

6.1 Preprocessing

Modelling and evaluation data cover the period of June 1–5, 2023, yielding approximately 13.6 million labelled pixels. Due to time restrictions, the month column is dropped. hour_of_day was cyclically encoded as sine and cosine components to preserve diurnal continuity across the hour 23–0 boundary. Missing values in aerosol_index (~0.2%) are imputed with the median to avoid data leakage. Extreme outliers in so2_no2_ratio and wind_x_ratio, which exhibit heavy-tailed distributions, are clipped at the 1st and 99th percentiles prior to any scaling, which may otherwise distort downstream scaling and degrade tree split quality. The dataset is heavily imbalanced, with volcanic pixels (label = 1) comprising approximately 11% of the 13.6 million observations. Rather than resampling, inverse-frequency class weights were computed on the training partition only, penalising minority-class misclassification without altering the data distribution.

6.2 Train/Test Splitting

An initial random pixel-level split produced artificially high accuracy, as pixels from the same emission source appeared in both training and test sets. We therefore adopt a source-level holdout strategy: pixels are grouped by nearest_source_name (754 distinct sources) and randomly partitioned 80/20, ensuring evaluation on geographically unseen sources.

6.3 Model Selection and Evaluation

Gradient Boosted Trees (GBT), Random Forest (RF), and Logistic Regression (LR) as a linear baseline are trained within PySpark MLlib pipelines. The GBT configuration follows Friedman [9]’s

Table 1: Feature descriptions and justifications for the SO₂ classification pipeline.

Feature Count	Feature	Justification
1	Nearest Source Name	Necessary to prevent data leakage by filtering pixel distribution 80/20 per source.
2	SO ₂ NO ₂ Ratio	Industrial emission is characterised by high SO ₂ and NO ₂ emission, whilst volcanic sources emit high SO ₂ with low NO ₂ [13]. Consequently, the expected ratio of these particulates should vary significantly depending on the source type.
3	Vertical Column (VC) Emission Height Ratio	Industrial emission is typically found at 1 km elevation whilst natural emissions are found at 7 km [24]. By using the vertical column as a proxy for pollutant density, the VC height ratio should possess high discriminative property.
4	Aerosol Index	Volcanoes are much higher producers of aerosols than power plants, specifically producing sulfate aerosols around volcanic eruption sites [18]. This means the aerosol index should increase in naturally emitted pollutants and act as an independent corroborating signal.
5	Wind Corrected Distance	The implementation of back trajectory analysis. The implementation of back trajectory analysis to estimate backward-in-time advection by 1 day, in line with the global average of SO ₂ lifespan [22].
6	Raw Distance	To benchmark the impact of the back trajectory analysis.
7	Source Elevation	Implements the pollutant height characteristics independent of vertical column mass.
8	Wind Speed	Higher wind speeds directly influence the distance plumes advect from source and can signal a noisier estimate of pixel source attribution.
9	Wind Speed and SO ₂ NO ₂ Concentration Ratio	An interaction feature between wind speed and the SO ₂ NO ₂ ratio. With high wind speeds, older plumes can enter the observation window inflating SO ₂ NO ₂ ratios due to the shorter lifespan of NO ₂ .
10	Cloud Fraction	Cloud cover is already mitigated by the QA pixel filtering, but its direct inclusion signifies the potential for noise in imagery.
11	Solar Zenith Angle	Higher solar zenith angles lead to longer paths between sun, atmosphere and satellite. As a result, they decrease surface irradiance and contrast causing atmospheric scattering — i.e. noise [11].
12	Sensor Zenith Angle	Higher sensor zenith angles lead to spectral interference and imperfect column measurements due to the offset angle of the satellite [22].
13	Surface Altitude	Directly signals volcanic overhead pixels which will hang over a higher surface area than industrial sources.
14	Month	A signal for seasonality, not relevant to our analysis as we use a 5-day sample from June but included for proof of concept.
15	Hour	Another proxy of solar zenith angle and diurnal activity of pollutants.
16	Label (Natural or Industrial)	Binary classification label.

stochastic boosting recommendations to reduce overfitting through shallow trees and row subsampling. RF used numTrees=100 and maxDepth=10, trading slightly deeper trees for ensemble diversity. LR required separate robust and standard scaling pipelines for

heavy-tailed and well-behaved features respectively. Due to computational constraints on the GCP Dataproc cluster, extensive hyperparameter tuning was not performed. Models were assessed on AUC-ROC, AUC-PR, F1, precision, recall, specificity, and Matthews Correlation Coefficient (MCC). AUC-PR and MCC are particularly informative under class imbalance [20]. An ablation experiment

comparing wind-corrected against raw pixel distance labels quantifies the contribution of ERA5 wind back-trajectory correction to classification performance.

6.4 Numerical Results

This section evaluates the three classifiers (GBT, RF and LR) on volcanic SO₂ source attribution, with three specific goals: identifying the best-performing model, comparing the pixel-level and source-level splitting and quantifying the contribution of ERA5 wind back-trajectory correction to classification performance. All models are evaluated on the June 2023 5-day pilot subset under source-level holdout. The wind-corrected dataset (about 13.6M pixels) and the raw pixel distance dataset (about 12.7M pixels) serve as the two experimental conditions. Logistic Regression serves as the linear baseline.

As shown in Figure 1, GBT and RF both achieve high accuracy (0.91) and F1 (0.93) under wind-corrected labels, substantially outperforming Logistic Regression (accuracy 0.80, F1 0.84). This shows a non-linear decision boundary. Under class imbalance, Saito and Rehmsmeier [20] demonstrate that AUC-PR and MCC are more informative than accuracy alone. The class imbalance can be more clearly seen in MCC: GBT and RF score 0.64 to 0.67 against LR's 0.44, indicating that LR's apparent accuracy is partly driven by the majority industrial class.

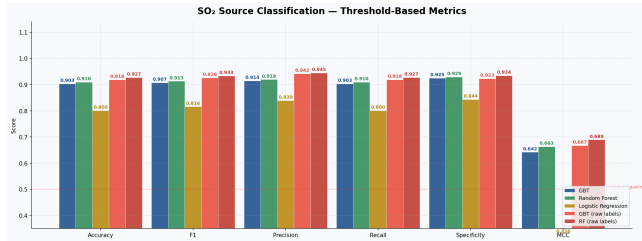


Figure 1: SO₂ Source Classification — Threshold-Based Metrics

Figure 2 shows the normalised confusion matrices for all three classifiers under source-level holdout. While pixel-level splitting yields near-perfect matrices, this constitutes data leakage which makes the model learn source identity rather than generalise to unseen emitter.

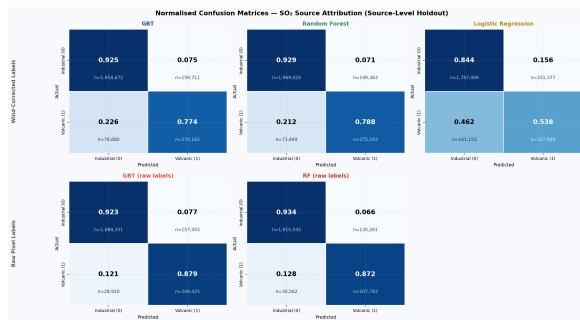


Figure 2: Normalised Confusion Matrices — SO₂ Source Attribution (Source-Level Holdout)

Figure 3 further reveals that the AUC-PR of LR collapses to 0.33 against 0.75–0.85 for tree models, confirming that tree-based models capture the non-linear structure of the classification boundary far more effectively.

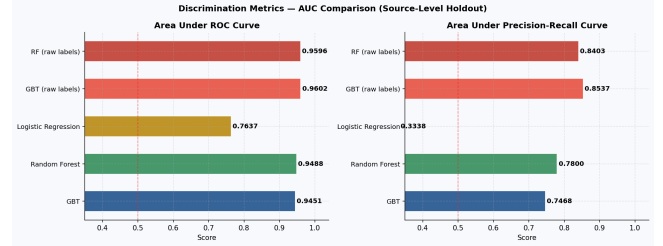


Figure 3: Discrimination Metrics — AUC Comparison (Source-Level Holdout)

For the wind correction contribution, Figure 4 demonstrates that wind-corrected labels consistently improve precision by 0.026–0.028 across both GBT and RF, reducing false volcanic attributions at a small recall cost.



Figure 4: End-to-End Wind Correction Effect (Source-Level)

Moreover, Figure 5 shows that wind_corrected_dist_km achieves higher feature importance than the equivalent raw pixel distance in both GBT and RF, confirming that ERA5 wind back-trajectory displacement is more informative than naive pixel proximity for distinguishing volcanic from industrial SO₂ sources.

7.4 Scaling Behaviour and Bottleneck Analysis

The two experiments can be interpreted using complementary scaling laws. Amdahl’s Law [2] explains the strong scaling results that when part of a program must run sequentially, adding more workers yields diminishing returns. In this pipeline, GBT training acts as the main bottleneck, since each tree depends on the errors from the previous one.

In contrast, Gustafson’s Law [10] explains the weak scaling behaviour. When the parallel portion of the workload grows with problem size, increasing worker count can keep runtime roughly constant. This pattern is reflected in the results, where normalised runtime stays close to 1.0 up to 8 workers before increasing modestly to 1.29× at 16 workers. The per-stage speedups in Figure 8

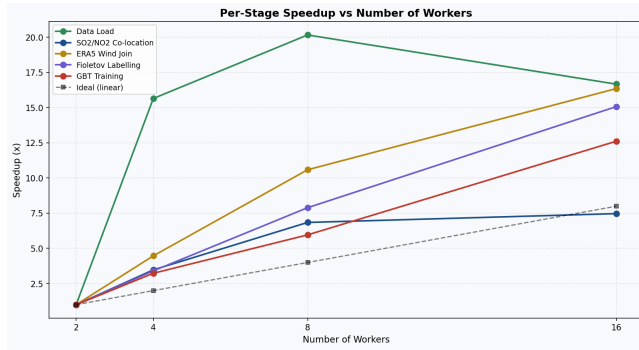


Figure 8: Per-stage speedup across worker configurations.

highlight clear differences across the pipeline. Data loading (S1) achieves the highest gains, reaching around 20× speedup at 8 workers, as additional workers allow more parallel reads from GCS. The ERA5 wind join (S3) scales close to linearly, achieving 16.3× speedup at 16 workers, since the workload partitions cleanly by geographic region with minimal communication. The Fioletov labelling stage (S4) also scales well, reaching 15.0× at 16 workers, aided by the small catalogue size which can be broadcast to all workers instead of shuffled. In contrast, GBT training (S5) shows the weakest scaling, reaching only 12.5× at 16 workers and consistently remaining below the ideal line, reflecting the sequential dependency between boosting iterations as described by Amdahl’s Law [2]. Overall, the pipeline demonstrates strong scalability as classification performance remains stable across all runs, with AUC-ROC between 0.9993 and 0.9994 and F1 scores between 0.9893 and 0.9897, indicating that increasing worker count does not affect model quality.

7.5 Partitioning Strategy

Shuffle partition count strongly affects shuffle-heavy stages, as shown in Figure 9. Under a fixed 400 partitions setting, the ERA5 wind join takes around 1,400 seconds at 2 workers, the highest cost of any stage, and decreases sharply as more workers reduce over-partitioning overhead. The SO₂/NO₂ co-location stage shows a similar pattern, starting at around 390 seconds.

Under proportional partitioning, both stages remain low and stable across all configurations. The ERA5 wind join stays between

28–43 seconds, while co-location remains between 62–74 seconds, regardless of worker count. This indicates that matching partition count to cluster size largely removes scheduling overhead. Overall, these results suggest that partition counts should scale with cluster size rather than remain fixed, especially for shuffle-intensive workloads.

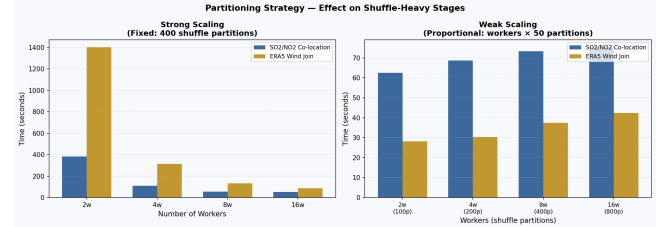


Figure 9: Effect of partitioning strategy on shuffle-heavy stage runtimes.

8 Conclusion

This report presented a distributed PySpark pipeline for classifying satellite-observed SO₂ emissions as volcanic or industrial in origin, deployed on GCP Dataproc across a five-day pilot dataset of 13.6 million globally distributed pixels. The pipeline integrates TROPOMI SO₂ and NO₂ observations with ERA5 850 hPa wind back-trajectory correction and the Fioletov et al. [7] source catalogue, enabling pixel-level classification at a scale that would be impractical with traditional ground-based monitoring approaches.

On classification performance, GBT and Random Forest both achieved accuracy of 0.91 and F1 of 0.93 under source-level holdout, substantially outperforming the Logistic Regression baseline and confirming a non-linear decision boundary. ERA5 wind correction consistently improved precision by 0.026 to 0.028 across both models, providing the first quantitative measurement of this effect in the literature and confirming that atmospheric displacement is a more informative signal than raw pixel proximity for source attribution.

On distributed performance, the pipeline achieved a 13.4× speedup scaling from 2 to 16 workers under strong scaling, reducing a near 7-hour workload to just over 30 minutes. Weak scaling showed only a 29% rise in normalised runtime for an 8-fold increase in data size, consistent with Gustafson’s Law [10]. The per-stage analysis identified GBT training as the main bottleneck under Amdahl’s Law [2], while the ERA5 wind join and Fioletov labelling stages scaled close to linearly. The partitioning strategy comparison further showed that scaling shuffle partitions with cluster size almost entirely removes scheduling overhead in shuffle-heavy stages.

Several limitations should be noted. The experiments used only a five-day pilot window from June 2023, and results may differ across seasons or years. Hyperparameter tuning was not performed at scale due to computational constraints. The pipeline also did not incorporate the full set of socioeconomic features originally proposed in the project, such as night-time light intensity and population density, which could improve classification in regions where volcanic and industrial sources are geographically close.

Future work could extend the pipeline to the full TROPOMI archive from 2018 to the present, enabling long-term analysis of

emission trends and better support for health policy and environmental regulatory compliance. Adding socioeconomic features would enrich the feature space and improve interpretability in policy-relevant contexts, as originally motivated. On the infrastructure side, training independent GBT models per geographic region in parallel could reduce the serial bottleneck identified in the scaling experiments. Extending the pipeline to near real-time processing using Spark Structured Streaming would make the system more practical for operational SO₂ monitoring.

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Use of Generative AI and Chat Logs

Member One Candidate Number: 72560

For this project, I used Claude for drafting and debugging the data ingestion preprocessing pipeline, for developing our approach to the project and for enhancing the report with comprehensiveness in detail. <https://claude.ai/share/78ee74f3-a3dc-49f0-a134-c3e9b39d4443>

Member Two Candidate Number: 69942

For this project, I used Claude for drafting and debugging PySpark and GCP file errors, and enhancing essay planning and distributed computing understands with 'intellectual sparring'.

<https://claude.ai/share/4f0503a3-ec81-4a97-a007-0e8469f99a9a>

Member Three Candidate Number: 69308

For this project, I use Claude for debugging the PySpark and Google Cloud errors, help me draft the difficult code and visualisation plots, help me understand the detail of task and polish my report.

<https://claude.ai/share/77b578b6-9072-4888-a03c-54a519889c60>

<https://claude.ai/share/14c100f1-62a0-466d-be63-40255d686955>

<https://claude.ai/share/676cff86-bccc-4f0a-b24e-22ff5fcb2a22>

<https://claude.ai/share/2b943d67-0d0f-4725-bf60-487700eb367d>

<https://claude.ai/share/07f4358d-df42-4d11-905e-d6914c492841>

<https://claude.ai/share/a6a496ba-3de6-4eca-8e16-763d2decde39>

Member Four Candidate Number: 61175

I used Claude for coding for strong and weak scaling and solve any technical issues in GCP.

<https://claude.ai/chat/b718e904-0430-4190-90e6-2f78932f2cc8>

Statement about individual contributions

The individual technical contributions of each group member is summarised in a separate pdf file which could be found on Github. The contribution of each group member towards all tasks is 25%. All contributions per task add to 100%.